

Beyond the Frame: A Constructivist Framework for Cinematic Education in Immersive VR Environment

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Figure 1: In the Clouds: Vancouver from Cypress Mountain.

ABSTRACT

Virtual reality (VR) is increasingly used in education because it can create embodied learning situations that are difficult to reproduce in physical classrooms. From a constructivist perspective, its value lies less in “presence” than in enabling learners to construct knowledge through situated experimentation. Building on prior VR learning research [3,4], we present CineSim VR, an experiential-learning environment that lets students enter a virtual film set to rehearse blocking, camera placement, and shot composition. In pilot workshops with media-production students ($n=50$), participants reported high engagement and that first-person, gesture-based interaction helped them link spatial composition with storytelling intent (Fig. 1).

Index terms: Virtual Reality, Cinematic Education, Constructivist learning, Experiential learning.

1 INTRODUCTION

Traditional film-production training is grounded in learning by doing on real sets. Students handle cameras and lights, adapt to on-set conditions, and refine decisions through rehearsal. Consistent with evidence that active learning improves student performance, effective teaching engages students in purposeful tasks that require them to make decisions and reflect on outcomes, rather than simply absorbing information [1]. Yet authentic set work requires substantial equipment, space, time, and manpower, which can limit how often learners can practise foundational skills. Our goal is not to replace on-set training, but to add a preparatory step. VR is chosen as the platform because it can simulate the spatial, embodied nature of filmmaking, such as camera position, blocking, and staging, while offering a safe, repeatable environment for low-cost

rehearsal and rapid iteration. CineSim VR lets students rehearse shot composition, blocking, and cinematography decisions in a virtual set while clarifying storytelling intent, with minimal manpower and equipment. This design is informed by experiential learning theory, which conceptualises learning as a process in which learners develop knowledge by reflecting on and acting upon experience [2]. The aim is to provide repeatable concrete experiences, support reflection on outcomes, and enable rapid iteration before students transfer insights to real productions in later studio sessions.

2 SYSTEM CONTRIBUTION AND NOVELTY

- Workflow (virtual director loop): Learners enter a film scene in first person and iteratively re-stage shots through rapid trial, reflection, and revision in the virtual set.
- Core interaction (low-friction controls): The system uses familiar VR teleport navigation and intuitive, direct manipulation (grab/drag-and-drop) for camera placement/movement and character blocking, allowing most users to begin within 5–10 minutes.
- Design stance (narrative immersion over photorealism): CineSim VR intentionally prioritises narrative immersion and embodied spatial reasoning over photorealistic fidelity, emphasising staging, blocking, and camera relationships as the basis for cinematic meaning-making and repeatable rehearsal.
- The reconstructed demo set used in the workshop is shown in Fig. 2.

3 EDUCATIONAL ACTIVITY

As part of the pilot evaluation, undergraduate students majoring in media production ($n=50$) participated on a voluntary basis. Students had prior exposure to basic video storytelling concepts. They were organised into 10 groups (~ five students per group) and joined a 75-minute workshop.

The activity centred on the bedroom scene from *The Graduate* (1967), which illustrates how camera placement and character positioning can communicate power relations and emotional tension through visual language. The session began with a 10-

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minute introduction to the film clip and its narrative intent, followed by a 5-minute briefing on CineSim VR controls and safety. Students then used Oculus Quest 3 headsets to enter the CineSim VR environment.

During the 30-minute practice-and-experiment phase, students first attempted to replicate key shots from the original scene and then explored alternative staging choices of their own design. In VR, they could re-stage character blocking and adjust camera angles and shot composition, allowing them to test how visual decisions reshape meaning. Staff provided on-site technical support to ensure smooth operation, while the learning focus remained on analysing and reconstructing relationships among characters, space, and shots.

The final 30 minutes were reserved for feedback collection. An example of the workshop workflow is shown in Fig. 3.



Figure 2: Demo Scene



Figure 3: Class Activity

4 USER STUDY AND EVALUATION

Method: Participants were undergraduate media-production students (n = 50) recruited from course workshops (voluntary participation). The workshop followed the shot-replication and iteration workflow described above. Data sources included (1) a post-session questionnaire (1–10 ratings) measuring perceived learning, engagement, and usability/interaction friction, and (2) a semi-structured focus group facilitated by an independent moderator was summarised into themes (e.g., spatial understanding, motivation, control difficulties).

To strengthen the credibility of the pilot evaluation, we adopted a mixed-methods approach, combining a post-session questionnaire (10-point agreement ratings) with focus-group interviews. This design helps triangulate learners’ overall trends (quantitative) with their explanations and concrete examples (qualitative). The questionnaire primarily captured students’ perceived usability and learning value of CineSim VR, including how learning in VR compared with more conventional approaches. Selected results are reported in Tab. 1 and were generally positive, suggesting that CineSim VR may function as a useful supplement to traditional set-based practice.

Student Ratings of VR Learning Experience (n=50) (1 = strongly disagree, 10 = strongly agree)	
Statement	Rating (1-10)
Using VR helped me understand cinematic techniques better than traditional methods.	8.3
Using VR made the learning process more immersive and exciting.	8.46
The interactive nature of VR made learning more enjoyable and immersive.	8.02

Table 1: Selected question results from questionnaires

To reduce potential bias and encourage open discussion, the focus group was facilitated by a third-party moderator. Qualitative responses largely echoed the questionnaire findings. Students described CineSim VR as immersive and motivating, and many felt the first-person, embodied perspective made learning about spatial relationships and shot design easier. One participant noted that it

felt “like I am actually on a movie set as the director,” but with lower cost and flexible access. At the same time, students reported usability and design issues, such as sensitive camera control and clunky character placement, suggesting that early interaction friction may raise cognitive load for users without VR experience. Overall, the pilot indicates CineSim VR’s potential as a pre-visualisation and rehearsal tool, while pointing to the need for stronger evaluation designs (e.g., comparison conditions and performance-based measures) to substantiate learning and transfer claims.

5 FUTURE DEVELOPMENT AND CONCLUSION

While the pilot study results suggest that VR can support film-learning processes such as spatial understanding, shot design, and blocking, adoption is not automatic. Some learners remain hesitant to try unfamiliar technologies, and early interaction friction can reduce confidence and distract from learning goals. Future development will therefore prioritise accessibility and ease of adoption alongside content expansion. First, the learning activity will be refined with a stronger onboarding flow, such as guided tutorials, clearer interaction cues, and more intuitive controls to lower cognitive load and help inexperienced users become comfortable quickly. By reducing operational effort, learners can devote more attention to experimentation and reflection, which are central to constructivist and experiential learning.

Second, we will strengthen reflection using established instructional supports. After each shot is captured, CineSim VR presents a small in-headset “Critique” overlay with fixed, rubric-aligned prompts (e.g., how camera height/distance communicates power, whether the focal subject is clear, and whether blocking/eyelines read as intended), alongside a simple shot-planning rubric (blocking clarity, eyeline consistency, motivation of camera move, emotional intent). They are used to inspire users’ to think and review of their shots. We will also leverage the existing storyboard export for debrief, enabling groups to compare iterations side-by-side and articulate the rationale behind key camera and staging decisions.

In conclusion, CineSim VR is not a replacement for conventional production training, but a scalable preparatory environment for rehearsing cinematic decision-making with minimal resources. Pilot feedback was generally positive, indicating strong engagement and perceived learning value. With improved usability, onboarding, and feedback support, VR-based pre-visualisation can provide a practical pathway for students to build film-language knowledge through iterative, experience-driven practice.

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